CODESPHERE HACKATHON 2026

- **Problem Statement Title- "Enhancing Road Safety with AI-based pothole detection"**
- **Theme - AI for Social Good**
- **Team Name – Xplorer's**
- **Team Leader Name – Snehal Hari Kolhe**
- **College Name - SSBT's COET Bambhori,Jalgaon.**

Computer Association for Creative Technocrats COMPACT

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Road Mitra

GCOE, JALGAON

Where did the idea come from?

While returning from college,We witnessed a rider's bike suddenly hit an unexpected pothole, shattering its headlight. The shock on his face made us question- who is truly responsible for this hazard? Government, Contractors or Public?

Uneven roads and potholes are not just an inconvenience; they lead to accidents, costly repairs, and even loss of innocent lives.

This incident sparked the creation of ROAD MITRA - so our team created an AI-driven solution for real-time pothole detection, ensuring accountability and safer roads for everyone.

PROPOSED SOLUTION :

AI-Powered Image Validation & Severity Detection
- ✔ Ensures accurate pothole reports by validating user images
- ✔ Categorizes issues into White, Yellow, and Red Zones for priority-based repair

Contractor Accountability & Transparency
- ✔ Strict deadlines & penalties ensure timely repairs
- ✔ Enhances road maintenance efficiency & government transparency

Gamified User Engagement & Rewards
- ✔ Encourages pothole reporting through a reward system
- ✔ Users earn redeemable vouchers, boosting community participation

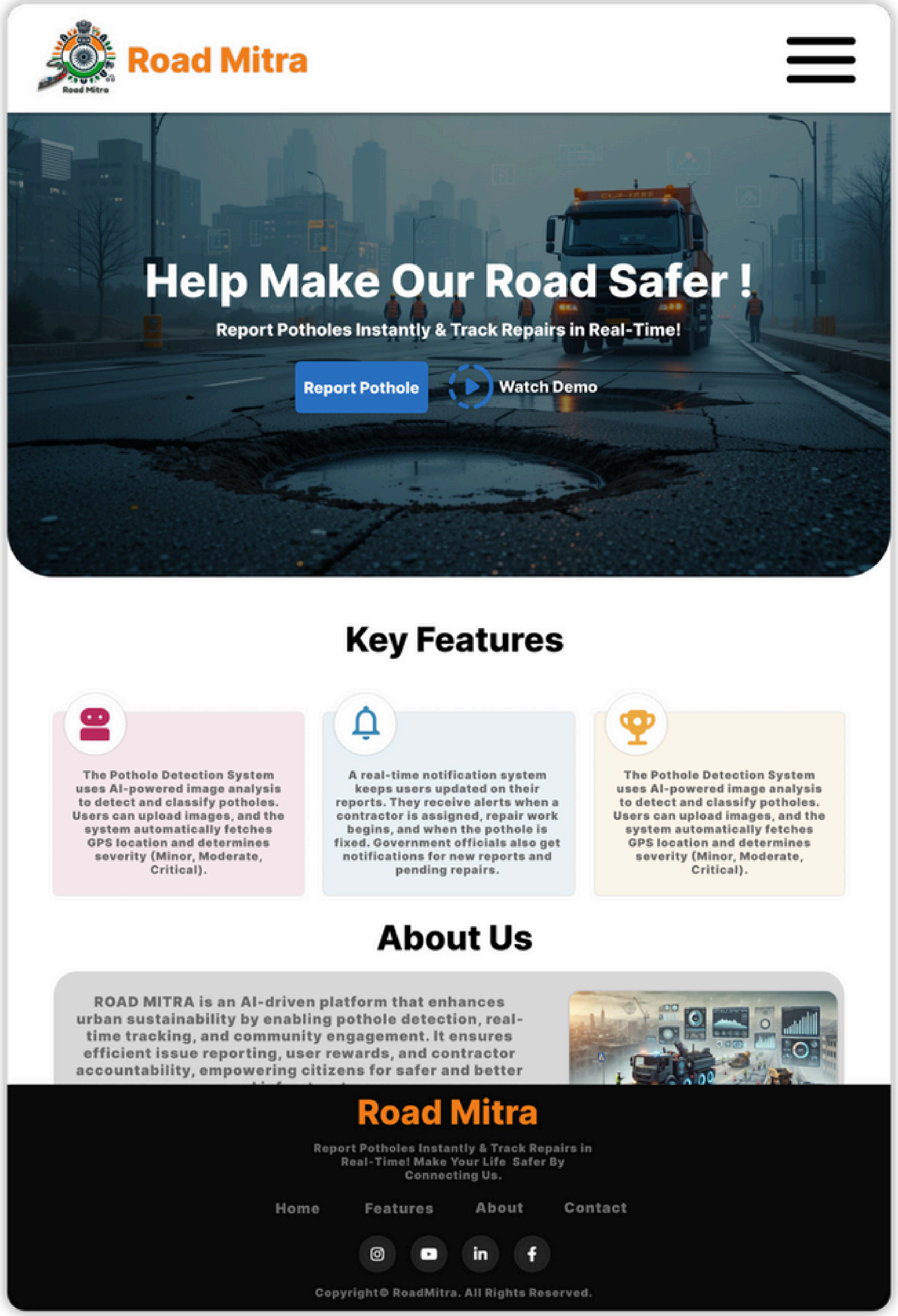
Real-Time Tracking & AI Chatbot Assistance
- ✔ Live updates on repair status for users
- ✔ AI-powered chatbot for easy complaint submissions & query resolution

What is Road Mitra?

ROAD MITRA is an **AI-driven** platform that enhances **urban sustainability** by enabling **pothole detection**, **real-time tracking, and community engagement**. It ensures **efficient issue reporting, user rewards**, and contractor accountability**, empowering citizens** for safer and **better road infrastructure.**

Features :

- AI-Powered Pothole Detection
- Gamified Reporting & Rewards
- Multilingual Support
- Real-Time Issue Tracking
- Chatbot Assistance
- Community Engagement
- Secure OTP Verification
- Instant Notifications & Alerts



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TECHNICAL APPROACH

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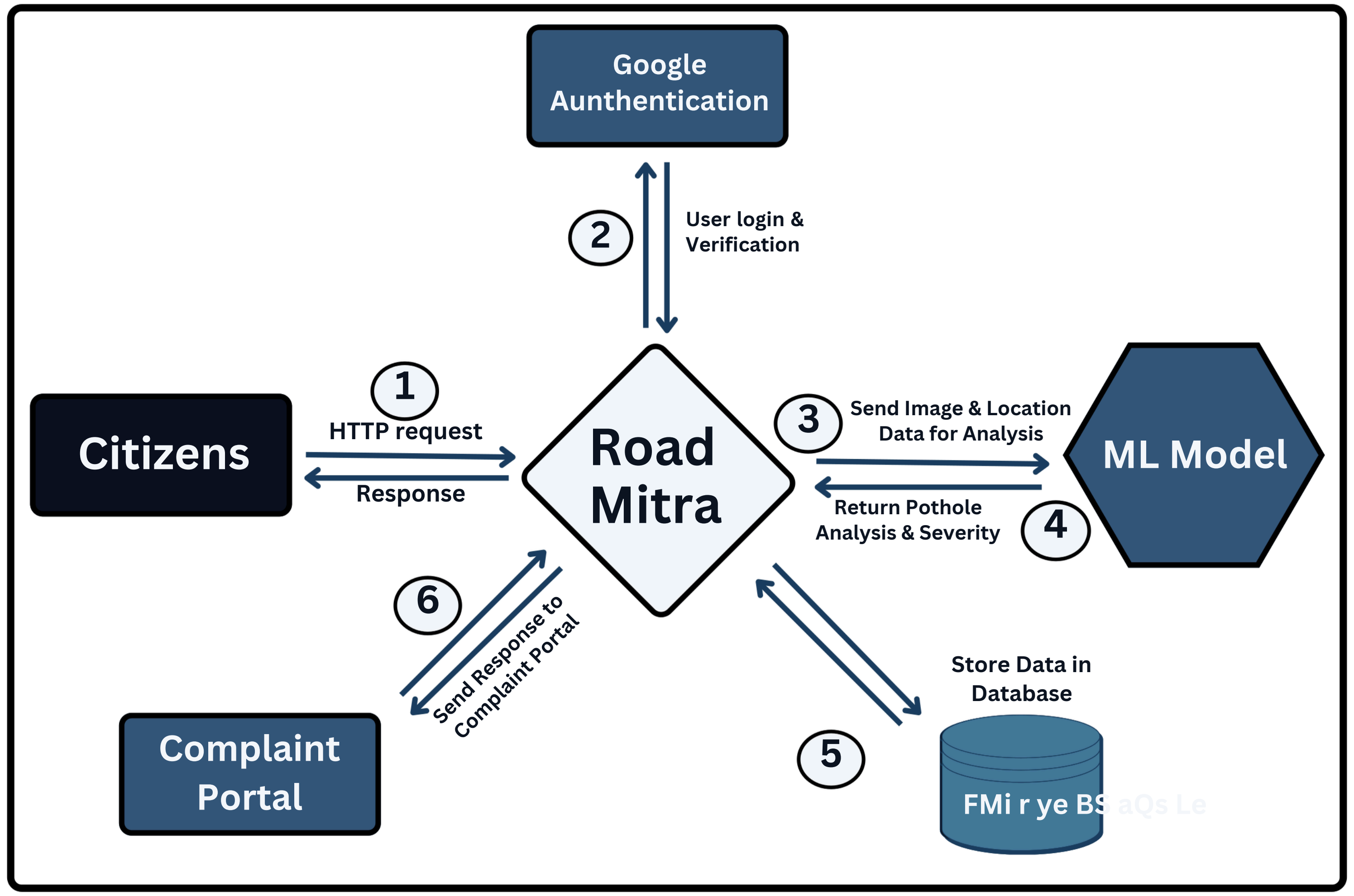

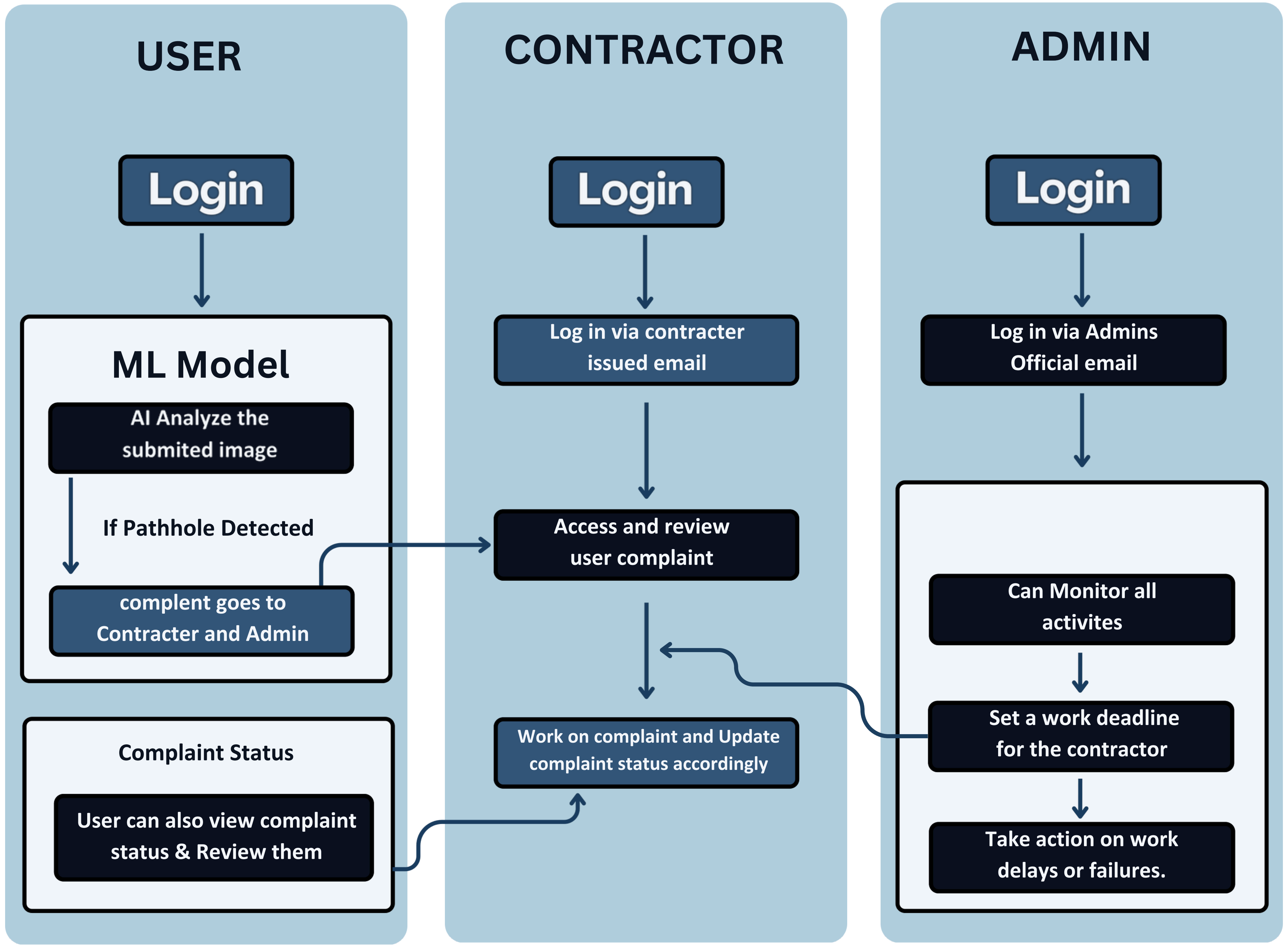

Tech Stack:

- **Frontend -** HTML5 , CSS3 , JavaScript
- **Backend -** Python (Django)
- **Database -** MySQL
- **AI/Ml -** OpenCV , Tenserflow
- **Notification -** Firebase Cloud Messaging

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FEASIBILITY AND VIABILITY

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- **User-Centric Innovation & Smart Infrastructure**
  - User-Friendly App
  - AI-Driven Efficiency
- **Strong Revenue Model**
  - Revenue Streams
  - Scalability & Smart Integration
- **User-friendly platform with AI Assistance**
  - Smart Reporting &Tracking
  - Instant Alerts & Multilingual Support
- **Corruption-Free, Transparent System**
  - AI-Powered Reporting
  - Real-Time Monitoring
- **Scalable to Multiple Cities**
  - SmartCity Integration
  - Optimized Repairs

CHALLENGES & SOLUTIONS :-

- **Fake or Irrelevant Reports** – Users may upload AI-generated or unrelated images.
  ✓Solution – Require real-time photo uploads with automatic geo-tagging for authenticity
- **User Engagement Drop –** Lack of motivation to report issues consistently. .
  ✓Solution – Reward users with points and redeemable vouchers to encourage participation
- **Communication Barriers –** Users may not receive timely updates on their reports.
  ✓Solution – Send email notifications and OTP verification for clear and transparent communication.
- **Loss of Trust in the System –** Delayed responses lead to user frustration and disengagement.
  ✓**Solution–**Set deadlines, send regular updates, and use AI to prioritize urgent fixes.
- **InefficientIssue Resolution –** Reports may not be addressed in a timely manner.
  ✓**Solution–**Implement AI-driven prioritization and smart tracking to ensure quick resolutions.

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IMPACT AND BENEFITS

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OUR AUDIENCE :

Common People:

- **Safer Roads,FewerAccidents –** Detecting and fixing potholes reduces road mishaps.
- **Smooth Driving Experience –** Less vehicle wear and tear, saving on repair costs.
- **Quick Reporting System –** Instantly report potholes with real-time tracking.

Contractor / worker :

- **EfficientRepairing –** Speeds uppothole detection and repair with real-time data.
- **Enhanced Safety –** Prevents accidents by fixing hazards quickly.
- **Transparent Work –** Ensures funds go to real repairs, reducing corruption.
- **Time & Cost Savings –** Optimized scheduling cuts costs and boosts efficiency.

Government :

- **Real-TimeMonitoring** – Live tracking boosts decision-making and accountability.
- **Reduced Corruption** – Digital records ensure transparency and prevent fund misuse.
- **Cost-Effective Management** – Data insights cut maintenance costs and extend road life.

- **Safer Roads, Fewer Accidents**
  Real-time pothole detection helps authorities act fast, preventing accidents.
- **Transparent Road Maintenance**
  Automated tracking prevents fund misuse, ensuring proper pothole repairs.
- **Swift & Efficient Repairs**
  Real-time alerts help contractors fix potholes faster, reducing delays.
- **Stronger Government Oversight**
  Real-time monitoring ensures accountability, better roads, and stricter enforcement.
- **Smoother & Safer Roads**
  Timely repairs enhance driving comfort, reduce damage, and improve safety.

"Road Mitra isn't just a project, it's a movement for safer roads and a corruption-free system!"

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RESEARCH AND REFERENCES

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- **Ministry of Road Transport and Highway**
  - Research into India's e-governance systems, such as the Ministry of Road Safety and the online grievance portal, provided a foundation for Road Mitra’s citizen-government interaction model.
  - Reference:https://app.powerbi.com/viewr=eyJrIjoiMjIzMTY5MmQtNjZmZC00OTAyLTkzOGMtYWEyOWYwZDE1YjU2IiwidCI6IjViN2ExMDIzLTI1ODgtNGU3Yi05MjZlLTgwYzllY2EwNWQ4OCIsImMiOjEwfQ%3D%3D
- **Public Issue Resolution**
  - Studiesonhowdigitalplatformsenhancepublicissueresolutionguidedthe"NotResolved"featureofSUDHAAR,ensuringuserfeedbackleads to effective government action.
  - Reference: Future of e-Government: An integrated conceptual framework - ScienceDirect
- **Reward Systems in Civic Apps**
  - Insights from civic apps like MyGov and FixMyStreet helped shape SUDHAAR's user reward system, providing incentives for continued user participation in reporting and tracking issues.
  - Reference:Gamification of Citizen Participation: A Systematic Mapping | IEEE Journals & Magazine | IEEE Xplore
- **Scalable Technology for Civic Apps**
  - **Next.js**, **Firebase**, and **AWS** were chosen for SUDHAAR's architecture to ensure a scalable, real-time, and secure platform capable of handling a large volume ofusercomplaints.
  - Reference: Real-World Next.js: Build scalable, high-performance, and modern web applications using Next.js, the React framework for production | Packt Publishing books | IEEE Xplore
- **AI and Mapping Integration**
  - **TensorFlow.js** and **Leaflet.js** were used for incorporating machine learning and geolocation, facilitating eco-friendly behavior tracking and precise issuereporting in theapp.
  - Reference: Artificial intelligence in marketing: Systematic review and future research direction - ScienceDirect